

Erdős Problem #942

EP-942

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $h(n)$ count the number of powerful (if $p \mid m$ then $p^2 \mid m$) integers in $[n^2, (n+1)^2)$. Estimate $h(n)$. In particular is there some constant $c > 0$ such that $h(n) < (\log n)^{c+o(1)}$ and, for infinitely many n , $h(n) > (\log n)^{c-o(1)}$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>